

Next Generation Underwater Breathing Apparatus for Explosive Ordnance Disposal Diving

PROBLEM STATEMENT

The MK 16 Underwater Breathing Apparatus (UBA), the primary rebreather for U.S. Navy Explosive Ordnance Disposal (EOD) divers, was first introduced in the 1980s and fielded in the 1990s. Despite its long service history, the system has received minimal, significant technological upgrades as it continues to meet most legacy mission requirements. However, this technological stagnation presents critical challenges to mission readiness, sustainability, and—most importantly, diver safety in the new missions and critical roles of the EOD diver.

- **Sustainability and Logistics:** As a legacy system, the MK 16 faces growing logistical and sustainability hurdles. Sourcing components for an aging system is inefficient and costly, impacting the overall readiness and availability of mission-critical equipment. The EOD Program Office is looking for a sustainable system with a larger user footprint to ensure long-term availability for future divers.
- **Modern Safety & Operational Requirements:** The MK 16's design lacks critical safety features now standard in modern diving equipment. The absence of predictive warnings exposes operators to preventable, life-threatening risks. Furthermore, ensuring the system meets all modern EOD operational requirements, including minimal acoustic and magnetic signatures, remains paramount for mission success.

This problem has become an urgent requirement to procure a sustainable, long-term replacement for the MK 16 UBA. This "Next Generation" system must be a military/commercially available or readily adaptable solution that incorporates essential technology upgrades and modernized features focused on enhancing diver safety. Key capabilities are listed below in primary, secondary and tertiary attributes.

PROPOSED SOLUTION

The proposed solution is the procurement and fielding of a Next Generation UBA to replace the MK 16 system.

This system will be a low influence signature (magnetic and acoustic), sustainable, long-term military or commercially available solution that integrates modern technology to maximize diver safety and operational effectiveness. The core of the solution is to equip EOD divers with a system that meets the demands of their evolving mission set. The system must be in conformance or able to be in conformance with current Navy Dive Tables.

A solution brief's Technical Merit and Relevance will be based on the ability to meet the primary attributes and the cumulative evaluation of the secondary and tertiary solution attributes/capabilities:

Primary Attributes:

- Closed-circuit, back-mounted (majority of system is on the back not affixed to front of diver), mixed-gas diving system that provides capability to a variety of military diving operations in a variety of ocean depths. The system requires the capability to operate with both N₂O₂ (air) and HeO₂ (88/12) diluents from depths 0 to 328 feet (100m) of seawater (fsw). Bottom time at 328 Ft (100m) should be a minimum of 20 minutes.
- Resistive Effort (WOB/L or KPa) in accordance with Technical Manual 15-01 Table 4-3, Limits in kPa (RE) (shown below).

Diluent	PPO ₂	Resistive Effort Limit (WOB/L or KPa)	RMV
HeO ₂ (88/12)	1.3 300 FSW to 328 FSW	2.34	62.5
HeO ₂ (88/12)	1.3 all depths to 300 FSW	2.41	62.5
N ₂ O ₂ (79/21) and HeO ₂ (88/12)	1.3 all depths to 150 FSW	2.04	62.5

Switchover at 32 FSW and 12 FSW on Ascent in accordance with table below.

Depth (FSW) +/- FSW	PPOs +/- 0.15ATA
DESCENT	
0-32	0.75
32-328	1.3
ASCENT	
328-12	1.30
12 - 0	0.75

- Meets requirements or capable of meeting requirements of STANAG 2897/Class A referencing STANAG AEODP-7/AMP-15 for signature certification.
- Compatible with a full-face mask (FFM) that meets the magnetic signature requirements of STANAG 2897/Class A as well as can meet all Approved for Navy Use (ANU) and SEA 00C requirements for use (FFM will be part of procurement and certification process).
- Provide the diver real-time dive profile guidance regarding allowable bottom time and decompression schedules using an algorithmic computation capability that conforms to existing dive table limits through continuous monitoring of relevant dive rig sensor data. A head-up display is required to relay current rig status to the diver real time. All components must still meet the signature requirements above.

- The Head-Up Display (HUD) is defined as a visual component integrated with or attached to the diver's face mask, positioned at eye level. The primary purpose of the HUD is to provide crucial operational and safety alerts. The system's complexity may vary, ranging from an advanced display providing real-time data to a simplified interface using a sequence of signals. Regardless of the method, the HUD must, at a minimum, deliver continuous updates on the UBA status and provide clear, immediate warnings to the diver about emergent conditions
- The system must be capable of alerting the diver to system emergency situations during operations.
- The dive rig shall be fully operational in extreme maritime environments, including water temperatures 34°F and greater, and withstand operations in high-particulate water, zero-visibility conditions, and currents up to 1 knot.
- Capable of being stored, transported and operated IAW relevant guidance MIL-STD 810 (current revision).
- Components, materials, or processes posing potential safety or health hazards, if identified, shall not be used in accordance with NAVSEA TS500-AU-SPN-010.
- Gas flasks used with the system shall be able to achieve and maintain all relevant certifications for storage and transportation CONUS and OCONUS. Gas flasks shall be protected while in use, and while stored, such that inadvertent scratching, denting, and other damage does not occur to negate the certification.

Secondary Attributes:

- The UBA shall be able to record critical dive parameters for recall during forensics and dive accident investigations.
- The UBA shall provide a minimum of 300 minutes of onboard oxygen when gas flasks are pressurized to a maximum 4,350 PSI.
- The UBA shall be able to interface and pass data to an independent secondary dive computer (e.g., wrist-mounted) to be referenced in the event of a primary computer or HUD failure. This secondary dive computer should be capable of meeting the same STANAG standards.
- If provided, the buoyancy compensator (BC) must use independent gas source (i.e., separate from onboard breathing gas) for inflation. It must have attachment points and pockets for ancillary equipment. It must have integrated weights for buoyancy control and must accommodate use with or without dry suit. Integrated weights must be firmly secured and be easily and quickly removable. It shall be a modular component; must be able to be removed.
 - If the BC is not provided with the system, the system must be compatible with the current or in-review BC design currently approved for EOD Diving use with the VSW Viper and MK 16 UBA.
- The UBA and all associated equipment should not be susceptible to, or sustain damage when subjected to, a radiated electric field. Additionally, the equipment must not be significantly impacted by a radiated electric field strength of 20 V/m in the following frequencies
 - 30 MHz to 512 MHz Vertical.

- 30 MHz to 512 MHz Horizontal
- The weight of the system-in-air, with all components, should not exceed 85lbs. The UBA should be in a range from neutral to 4 pounds negatively buoyant with no more than 1.8 liters of gas in the breathing loop when tested
- For all electrical/computer components, the system shall be, with vendor assistance and utilizing the Government's Information Systems Security Management Team, able to achieve authority to operate.

Tertiary Attributes:

- The HUD associated should be available for future product improvements to include gathering and displaying data from other diving handheld and unmanned assets.
- Pre-Dive Setup is the setup and calibration and/or ppO2 setting adjustment conducted before a dive. Setup time shall not exceed 30 minutes.
- The UBA shall be compatible with, or the vendor shall provide a plan for future implementation of, a safe diver egress that will provide a breathable gas mixture in the event of UBA failure to egress the diver from maximum depth, without the need to doff the mask
 - Condition 1: Emergency Breathing System (EBS). The EBS is a UBA or UBA-similar system that is placed in the water column to be used in emergency situations that require the diver to have an alternate closed-circuit UBA. This UBA shall be capable of being staged in the water column in an operational state and be 100% ready to use when and if needed.
 - Condition 2: Emergency Gas System (EGS). The EGS (also known as a bailout) shall be independent of the UBA breathing loop with the capability to be switched to in emergency operation to give sufficient duration to either reach the surface for no decompression dives or reach the EBS during decompression profiles

AREA OF INTEREST (AOI) CRITERIA

At a minimum, the proposed solution must demonstrate the following criteria in addition to the technical attributes:

- Relevance - Is the solution relevant to the AOI?
- Merit - Does the solution have technical merit?
- Uniqueness - Is the solution unique, underutilized, and/or innovative and compelling?
- Viability - Is the company viable?
- Rough Order of Magnitude (ROM) - Is the ROM (Price Estimate) reasonable?
- Schedule - Is the notional schedule acceptable based on the AOI requirements?
- Data Rights - Are the data rights the company is asserting acceptable to the Government?

SUBMISSION GUIDANCE

An individual vendor or team of vendors should not submit multiple solution briefs to the area of interest (AOI), unless each individual solution is substantively unique. If submitting as a team, the first page of the solution brief should list each company on the team and a primary point of contact for each company on the team as well as the lead vendor and point of contact for the overall team.

NOTE: In accordance with NSWC, Panama City Division (NSWC PCD) Commercial Solutions Opening (CSO) announcement number N61331-26-S-NQ17: The government reserves the right and ability to facilitate introductions of vendors with complementary capabilities. If teaming introductions by the government are provided, a vendor is not obligated to act on the recommendation. If a vendor elects to move forward with a government-recommended teaming arrangement, a Phase II pitch deck will be requested by NSWC PCD to facilitate a complete evaluation of the jointly proposed solution.

Pitch solutions will require more in-depth data being provided to include:

- 1) level of testing completed (independent and vendor) to date that will be available within 30 days of award for OT or FAR-based contract,
- 2) a detailed technical data package with drawings, schematics, cyber and system architecture to present and discuss data rights available if production moves forward,
- 3) Bill of Materials for an as-built configuration with materials noted for each (for hazard and off-gassing analysis),
- 4) Training, Fielding and Sustainment pathways,
- 5) key personnel and qualifications/company history within the military and technical UBA field, and
- 6) a production tour (onsite at manufacturing) and a technology readiness level assessment of meeting the production quantities and timeline.

FOLLOW-ON PRODUCTION

Companies are advised that any resulting awards from this AOI may result in the award of a follow-on, full-rate production FAR-based contract or other transaction without the use of further competitive procedures. The follow-on production contract or transaction will be available for use by one or more organizations in the Department of War and, as a result, the magnitude of the follow-on production contract or transaction could be significantly larger than that of the initial prototype acquisition. As such, any prototype acquisition will include the following statement relative to the potential for follow-on production: “In accordance with 10 U.S.C. 4022(f), and upon a determination that the prototype project for this transaction has been successfully completed, this competitively awarded prototype OT may result in the award of a follow-on production contract or transaction without the use of competitive procedures.”

ELIGIBILITY REQUIREMENTS

This solicitation will be awarded in accordance with the CSO process detailed within announcement number N61331-26-S-NQ17 NSWC PCD CSO to SAM.gov. In accordance with 10 U.S.C. 4022(d), at least one of the following conditions must be satisfied to enter into a prototype OT agreement: (A) at least one nontraditional defense contractor or nonprofit research institution participating to a significant extent, (B) all significant participants are small businesses or nontraditional defense contractors, or (C) at least one third of the total cost of the prototype project is to be paid out of funds provided by sources other than the Federal Government.